
The Single-Seed Noise-Floor Trap and an FFN-Capacity Optimum in Fixed-Wall-Clock LLM Pretraining

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Abstract

We report rounds 6–10 (exp036–exp070) of an autonomous hyperparameter campaign on a fresh Karpathy autoresearch baseline (nanochat-style GPT, $\sim 50\text{M}$ parameters, depth 8, sliding-window attention, Muon+AdamW) under a fixed 5-minute wall-clock budget on a single H200, continuing from the $\text{TOTAL_BATCH_SIZE} = 2^{18}$ reference adopted in paper 001 (3-seed base mean $\text{val_bpb} = 0.97966$, $\sigma \approx 0.0015$). Two findings survive. First, a cautionary null: re-tuning learning rates at the new batch size produced two single-seed pilots that looked $\sim 1.5\sigma$ better than base (matrix_lr 0.03: 0.977569; embedding_lr 0.8: 0.977905), yet 3-seed paired confirmation returned both to the base mean (0.97968 and 0.97981) and their combination was worse (0.98025) — a clean demonstration that single-seed deltas below $\sim 2\sigma$ are noise, and that only multi-seed paired confirmation is trustworthy. Second, a confirmed structural lever: widening the FFN from the stock $4\times$ to $6\times$ improves val_bpb to a 3-seed mean of 0.97696 (best 0.975010), a paired $\Delta = -0.0027$ on all three seeds (paired $t \approx 4.3$), despite reducing step count from ~ 2000 to ~ 1720 . Ratios $7\times$ – $10\times$ do not beat $6\times$ ($8\times$ 3-seed 0.97763; $10\times$ 0.97985), locating a capacity optimum. This refines paper 001’s “update-count-limited” claim into a capacity $\times$ update trade-off: at this frame the model was FFN-capacity-limited, and capacity-per-step can beat step count. Net campaign state: $\text{val_bpb} = 0.975$ best, -0.055 versus the naive 1.0301 baseline.

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1. Introduction

Paper 001 of this campaign (OPHIS, 2026) established two results under a fixed 5-minute wall-clock budget: the cold-compile first run is an invalid control (F1), and halving TOTAL_BATCH_SIZE to 2^{18} tokens — doubling optimizer updates — was the sole change to clear the pre-registered 3-seed 2σ gate (F2). That paper closed by hypothesizing the frame is update-count-limited, and ranked as next steps (a) re-tuning learning rates at the new batch size and (b) structural exploration of new mechanisms.

Rounds 6–10 executed both, and each produced a finding:

1. F3 (Section 3): the learning-rate re-tune produced textbook noise-floor traps. Two single-seed pilots sat $\sim 1.5\sigma$ below the base mean and would have been adopted by any greedy single-seed policy; both evaporated exactly to the base mean under 3-seed paired confirmation, and their combination was worse than base. Sub- 2σ single-seed deltas carry no information at this frame.
2. F4 (Section 4): structural exploration found the campaign’s second confirmed lever. Widening the FFN hidden ratio from $4\times$ to $6\times$ improves val_bpb by a paired -0.0027 on all three seeds despite completing $\sim 14\%$ fewer steps. The frame is therefore not purely update-count-limited: near the default architecture it was FFN-capacity-limited, and the correct picture is a capacity $\times$ update product with an interior optimum.

Together the findings sharpen both halves of the campaign’s methodology: F3 hardens the statistical gate, and F4 corrects the mechanistic story that gate is protecting.

2. Experimental Setup

The frame, model, metric, and discipline are identical to paper 001 (OPHIS, 2026), which we summa-

size only briefly. Each run trains a nanochat-style GPT (Karpathy, 2025) of $\sim 50\text{M}$ parameters (depth 8, sliding-window attention, Muon (Jordan et al., 2024) + AdamW) for exactly 5 minutes of wall clock on a single H200; the metric is validation bits per byte (val_bpb, lower is better) at the end of the budget. Steps completed are an outcome, not a control.

Reference. All comparisons in this paper are against the configuration adopted in paper 001: `TOTAL_BATCH_SIZE = 218` tokens (~ 2000 steps), with a 3-seed base mean of `val_bpb = 0.97966` and single-seed noise $\sigma \approx 0.0015$.

Keep/discard gate. Unchanged: single-seed runs are screening pilots only; a change is adopted only if a 3-seed paired comparison against concurrent base controls clears 2σ . Rounds 6–7 stress-tested this gate (Section 3); it held.

3. The Single-Seed Noise-Floor Trap (Finding F3)

Paper 001’s top-ranked follow-up was to re-tune per-group learning rates at the new, smaller batch size, since batch-size/learning-rate coupling predicts the optimum may shift (McCandlish et al., 2018). Rounds 6–7 swept `matrix_lr` and `embedding_lr` around their defaults at 2^{18} .

Two pilots flashed apparent signal. As single seeds, `matrix_lr = 0.03` reached 0.977569 and `embedding_lr = 0.8` reached 0.977905 — respectively -0.0021 and -0.0018 below the base mean, i.e. $\sim 1.5\sigma$, the most promising-looking pilots of the campaign to that point. A greedy single-seed policy adopts both.

Three-seed paired confirmation erased both effects exactly:

- `matrix_lr 0.03:` seeds 0.977569/0.982521/0.978951, mean **0.97968** — indistinguishable from the base mean 0.97966. Rejected.
- `embedding_lr 0.8:` seeds 0.977905/0.981219/0.980303, mean **0.97981** — indistinguishable from base. Rejected.
- Their combination (`matrix_lr 0.03 + embedding_lr 0.8`): mean **0.98025** — worse than base. Rejected.

The pilots were not weak versions of a real effect; they were the favorable tail of the seed distribution. Both

3-seed means landed within 0.00015 of the base mean, which is what a true effect size of zero predicts. The combination result is the sharper warning: stacking two “promising” noise draws produced a configuration mildly worse than doing nothing.

We state the operational lesson as F3: at $\sigma \approx 0.0015$, single-seed deltas below $\sim 2\sigma$ (~ 0.003 val_bpb) carry no adoption-relevant information; they select seeds, not configurations. Only $n \geq 3$ paired confirmation is trustworthy, and apparent pilot gains should be presumed to be noise until confirmed. This is the same statistical reality documented for pipeline-level variance elsewhere (Bouthillier et al., 2021), encountered here in the wild by an autonomous agent whose gate was designed for exactly this failure mode. The remaining learning-rate pilot (`matrix_lr 0.02: 0.977884`, $n=1$) was logged as noise without confirmation, closing the knob re-tune: the learning-rate surface at 2^{18} is flat, as it was at 2^{19} .

4. An FFN-Capacity Lever (Finding F4)

With scalar knobs exhausted on both batch sizes, round 8 screened structural mechanisms, one seed each: GeLU activation in place of ReLU² (0.99595), head dimension 64 (0.98646), rotary base 10^5 (0.98085), logit softcap 30 (0.98437), and a narrower FFN ratio of $3\times$ (0.98449) — all worse; an all-short attention-window pattern SSSS (0.97821) — neutral. The exception was widening the FFN hidden ratio from the stock $4\times$ to $6\times$: `val_bpb = 0.97501`, a single-seed -0.0047 , the first pilot of the campaign to exceed the 2σ screening bar and the best value observed to date.

Round 9 confirmed it. Three seeds gave 0.975010/0.978757/0.977104, mean **0.97696**, and the paired comparison against the concurrent base seeds improved on all three pairs with mean $\Delta = -0.0027$ (paired $t \approx 4.3$). Adopted — the campaign’s second confirmed lever and second milestone.

The mechanistically interesting fact is the step count. The $6\times$ FFN is slower per step: it completes ~ 1720 steps in the budget versus ~ 2000 for base, surrendering $\sim 14\%$ of its optimizer updates, and winning anyway. This directly refines paper 001’s Hypothesis H1. The frame is not update-count-limited simpliciter; `val_bpb` behaves as a product of per-step capacity and update count, and at the stock $4\times$ ratio the FFN — the transformer’s primary store of learned key–value memories (Geva et al., 2021) — was the binding constraint. We state the revised hypothesis:

Hypothesis H2 (capacity $\times$ update trade-off). At the 5-minute/1-H200 frame, val_bpb is set by a capacity $\times$ update-count product: widening the FFN buys per-step modeling capacity at a throughput cost, and the stock ratio $4\times$ is under-provisioned — the marginal val_bpb value of FFN capacity exceeds that of the steps it displaces — with the optimum near $6\times$, beyond which the throughput cost dominates.

Confidence: 0.75.

Both flanks were tested in rounds 9–10. Below: $3\times$ regresses badly (0.98449) despite more steps (~ 2130). Above: $8\times$ reached a 3-seed mean of 0.97763 with high seed variance (spread 0.974736–0.98065) and ~ 1530 steps — not better than $6\times$; $7\times$ as a single seed gave 0.97491, statistically indistinguishable from the $6\times$ seed-42 value (0.97501) and, per F3, not adoptable on one seed; $10\times$ gave 0.97985 at only 1367 steps, erasing the gain entirely. The ratio is settled at $6\times$ on parsimony: no wider ratio cleared the gate against it.

Interaction checks all came back negative. $6\times$ +SSSS (0.976579/0.979096) matches $6\times$ alone — the neutral window pattern adds nothing. $6\times$ +GQA with halved KV heads (0.983354/0.983825) trades quality for throughput the frame does not repay. Critically, $6\times$ at TOTAL_BATCH 2^{17} (0.980407/0.983063) is worse than $6\times$ at 2^{18} : the batch-size optimum of paper 001 is stable under the architecture change, consistent with the critical-batch-size picture (McCandlish et al., 2018) being governed by gradient noise rather than by FFN width. The confidence on H2 is held at 0.75 rather than higher because the $7\times$ flank is unresolved at $n=1$, the $8\times$ variance is anomalously high and unexplained, and “capacity” has only been probed through one axis (FFN width) — depth and aspect-ratio probes in paper 001 were losers, so the product form of H2 is an extrapolation from a single confirmed lever.

5. Results

Table 1 summarizes rounds 6–10. Δ val_bpb is reported against the round-6 reference mean 0.97966 (TOTAL_BATCH 2^{18} , 3-seed); negative is better. Data: campaign_log.jsonl, exp036–exp070.

Three features stand out. First, the only two sub-base 3-seed means in the table are the FFN $6\times$ family; every scalar-knob row collapses onto the reference under confirmation. Second, the FFN column is non-monotonic exactly as H2 predicts, with a regression on both flanks ($3\times$ and $10\times$) bracketing the $6\times$ optimum. Third, every combination built on top of $6\times$

(window pattern, GQA, smaller batch) is neutral or worse — the adopted configuration is a local optimum in the directions probed, not a springboard for further stacking.

6. Threats to Validity

1. Single-seed nulls. As in paper 001, most rejections in Table 1 rest on one seed against $\sigma \approx 0.0015$; “worse, discard” rows with $|\Delta| < 0.003$ should be read as “no effect resolvable at $n=1$.” F3 itself demonstrates how misleading such rows can be in the favorable direction; the same applies symmetrically.
2. The $7\times$ flank is open. FFN $7\times$ matched $6\times$ on its single seed. The settled ratio of $6\times$ rests on parsimony plus the $8\times/10\times$ regressions, not on a 3-seed rejection of $7\times$; the true optimum could sit between $6\times$ and $8\times$.
3. Unexplained $8\times$ variance. The $8\times$ seed spread (0.974736–0.98065) is roughly double the base σ . If wider FFNs genuinely inflate seed variance, the noise floor — and hence the gate — is configuration-dependent, which the current protocol does not model.
4. σ reused across configurations. The $\sigma \approx 0.0015$ estimate comes from base-configuration seeds; all z -style claims inherit it. The paired $t \approx 4.3$ for F4 is the more robust statistic, since it uses within-pair differences.
5. Capacity probed on one axis. H2’s “capacity” is operationalized solely as FFN width. Depth and aspect-ratio changes were losers in paper 001 under the same frame, so the product framing may be specific to FFN capacity rather than capacity generally.
6. Shared hardware. The node is shared; co-tenant load shifts step counts between waves. Concurrent controls mitigate but do not eliminate this.

7. Conclusion and Future Work

Rounds 6–10 delivered one hardened method and one adopted change. Methodologically, F3 shows the pre-registered gate doing exactly its job: two $\sim 1.5\sigma$ single-seed learning-rate “gains” at the new batch size collapsed onto the base mean under 3-seed paired confirmation (0.97968, 0.97981 vs. base 0.97966), and their combination was worse (0.98025); no single-seed delta below $\sim 2\sigma$ will be trusted for the remainder of the campaign. Substantively, F4 widens the FFN

Table 1. All configurations, rounds 6–10. Δ val_bpb is versus the 2^{18} 3-seed reference mean 0.97966 (negative = better). Single-seed rows are screening pilots; only $n \geq 3$ rows are eligible for the 2σ keep gate. The paired rows for the two learning-rate pilots illustrate F3: apparent $\sim 1.5\sigma$ single-seed gains that vanish under confirmation.

Config	mean val_bpb	Δ vs. reference	Seeds	Verdict
Reference: TOTAL_BATCH 2^{18} , FFN $4\times$ (~ 2000 steps)	0.97966	—	3	reference (pap
matrix_lr 0.03 (pilot)	0.977569	-0.0021 ($\sim 1.5\sigma$)	1	noise-floor tra
matrix_lr 0.03 (confirmation)	0.97968	$+0.0000$	3	reject — $\equiv$ ba
embedding_lr 0.8 (pilot)	0.977905	-0.0018 ($\sim 1.5\sigma$)	1	noise-floor tra
embedding_lr 0.8 (confirmation)	0.97981	$+0.0002$	3	reject — $\equiv$ ba
matrix_lr 0.03 + embedding_lr 0.8	0.98025	$+0.0006$	2	reject — wors
matrix_lr 0.02	0.977884	-0.0018	1	noise, not con
GeLU activation (vs. ReLU ²)	0.99595	$+0.0163$	1	worse, discard
head_dim 64 (8 heads)	0.98646	$+0.0068$	1	worse, discard
rotary base 10^5	0.98085	$+0.0012$	1	worse, discard
logit softcap 30	0.98437	$+0.0047$	1	worse, discard
attention window SSSS (all-short)	0.97821	-0.0015	1	neutral, discar
FFN ratio $3\times$ (~ 2130 steps)	0.98449	$+0.0048$	1	worse, discard
FFN ratio $6\times$ (~ 1720 steps)	0.97696 (best 0.975010)	-0.0027 (paired $t \approx 4.3$)	3	keep — adopt
FFN ratio $7\times$	0.97491	-0.0047	1	$\equiv 6\times$ at $n=1$;
FFN ratio $8\times$ (~ 1530 steps)	0.97763	-0.0020 , high variance	3	not better tha
FFN ratio $10\times$ (1367 steps)	0.97985	$+0.0002$	1	worse, discard
FFN $6\times$ + SSSS	0.97784	$\equiv 6\times$	2	no added gain
FFN $6\times$ + GQA (KV heads /2)	0.98359 / 0.98383	$+0.0041$	2	worse, discard
FFN $6\times$ + TOTAL_BATCH 2^{17}	0.98040 / 0.98306	$+0.0021$	2	worse; batch o

from $4\times$ to $6\times$ for a confirmed paired $\Delta = -0.0027$ on all three seeds ($t \approx 4.3$, mean 0.97696, best 0.975010) while losing $\sim 14\%$ of optimizer steps — refining paper 001’s update-count-limited picture into the capacity $\times$ update trade-off of H2 (confidence 0.75). The batch-size optimum of 2^{18} is stable under the new architecture, and ratios $7\times$ – $10\times$ do not beat $6\times$.

Net campaign state after ten rounds: best val_bpb = 0.975, an improvement of -0.055 versus the naive first-run baseline of 1.0301, with two adopted changes (TOTAL_BATCH 2^{18} ; FFN $6\times$) and two methodological findings (compile-cache confound; noise-floor trap) each backed by concurrent controls.

Ranked next experiments: (1) explore qualitatively new mechanisms rather than knob ranges — foremost n-gram/Engram-style memory modules, which add capacity through a different channel than FFN width and directly test whether H2’s product form generalizes; (2) resolve the $6\times$ – $8\times$ flank with 3-seed runs at $7\times$ if and only if a new mechanism does not dominate the queue; (3) estimate configuration-dependent seed variance (3 seeds at $8\times$ vs. base) to check whether the gate’s σ should scale with FFN width.

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